

1 **Feasibility of Field-Derived High-Resolution Event Replay for Traffic-**
2 **Signal Controller Testing**

3 **Shawn Strasser, P.E.**
4 Traffic Signal Operations Engineer
5 Oregon Department of Transportation
6 shawn.strasser12@gmail.com

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2 **Objectives.** Manual controller testing cannot reproduce everything a signal controller experiences over a day, including
3 interacting vehicle, pedestrian, coordination, and preemption activity. This study evaluates whether high-resolution
4 events recorded at field controllers can be replayed to test controllers and whether the resulting output sequences can
5 be aligned and compared automatically to identify operational differences.

6 **Methods.** An open-source Python package converts high-resolution logs to National Transportation Communications
7 for Intelligent Transportation Systems Protocol (NTCIP) calls and replays them to test controllers. Output events are
8 grouped by timestamp and aligned using dynamic time warping with Jaccard distance between event sets. It also
9 checks clearance intervals and incompatible signal phases and overlaps. Approximately 23 hours of events from each
10 of 25 production configurations were replayed to controller emulators. Separate runs tested repeatability, known timing
11 changes, and practical use for software-version acceptance.

12 **Findings.** Controller outputs from separate unchanged runs aligned with 99.6 percent mean sequence match and 97.7
13 percent mean timing match, demonstrating that controller responses to field-derived replay were deterministic enough
14 for automatic comparison. The method flagged operational differences in 11 of 14 configurations with deliberate
15 overlap clearance-setting changes and none of the 11 unchanged configurations. Applied to a software update, it also
16 identified real differences within otherwise similar full-day operation, including correction of a rail-preemption exit
17 bug. No incompatible signal phases or overlaps were detected.

18 **Novelty.** The study combines long-duration field-derived input replay with automatic alignment and comparison of
19 complete controller-output sequences across numerous production timing configurations.

20 **Practical Applications.** The open-source workflow substantially expands acceptance testing beyond manually tog-
21 gling individual inputs. Agencies can replay realistic signal activity across many configurations, automatically identify
22 unexpected differences before field deployment, reproduce field failures, and test software or timing corrections against
23 the same inputs.

1 INTRODUCTION

2 Controller acceptance testing is commonly limited by staff time and by the inputs staff can reasonably operate by
3 hand. Manually toggling detector, pedestrian, and preemption inputs remains useful for checking individual functions,
4 but it cannot reproduce a full day of interacting calls, coordination transitions, and preemption activity across many
5 production timing configurations.

6 This project uses field operation as the test sequence. Approximately 23 hours of high-resolution events from each
7 production controller are converted to National Transportation Communications for Intelligent Transportation Systems
8 Protocol (NTCIP) calls and replayed to test controllers. The resulting output events are compared with an accepted
9 field-derived baseline. The central question is:

10 Can high-resolution field events be replayed to test controllers, and can the resulting output sequences be aligned
11 and compared automatically to identify operational differences?

12 Feasibility requires both repeatability and sensitivity to change. Replaying the same inputs must produce controller
13 outputs similar enough to align reliably, while actual changes in signal operation must remain visible. The comparison
14 must also direct staff to important differences in signal phases, clearances, pedestrian service, preemption, coordina-
15 tion, and incompatible movements without requiring them to inspect millions of event records.

16 The evaluation covers 25 production configurations and three emulator campaigns using MAXTIME controller
17 software: version 2.15.1, version 2.18.1, and version 2.18.1 after overlap clearance settings were changed to prevent
18 occasional truncated yellow and red intervals. The primary contribution is evidence that field-derived controller
19 outputs were repeatable enough for automated comparison and that the method successfully identified real operational
20 differences. The specific software versions provide the case study; the finding is that this form of testing works.

21 RELATED WORK

22 Li et al. and the Idaho Transportation Department developed automated controller testing that used XML scripts to
23 activate inputs and verify predefined responses (Li et al. 2008; Ahmed et al. 2010). Tung extended NTCIP-based
24 automated testing across controller devices (Tung 2012; 2015). These systems established the value of repeatable
25 automated tests. The present method differs by deriving long-duration tests from field logs and using a prior complete
26 output trace as the expected response rather than requiring staff to author every input and assertion.

27 Controller-in-the-loop studies have connected controllers and emulators to traffic simulation through physical or
28 virtual interfaces (Wang et al. 2021). Stevanovic, Klanac, and Radivojevic evaluated high-resolution logging across
29 six controller vendors (Stevanovic et al. 2017). Those studies support the use of controlled controller environments
30 and show that output-event semantics require platform-specific validation. They did not evaluate field-log replay for
31 software regression across production configurations.

32 High-resolution event records provide tenth-second controller and detector histories (Sturdevant et al. 2012). DTW
33 provides a monotone alignment for sequences with local timing variation (Sakoe and Chiba 1978). The literature
34 reviewed for this study did not identify a prior evaluation combining full-day field-derived input replay, automatic
35 complete-output alignment, and repeated software and timing tests across numerous production configurations.

36 REPLAY AND TEST ARCHITECTURE

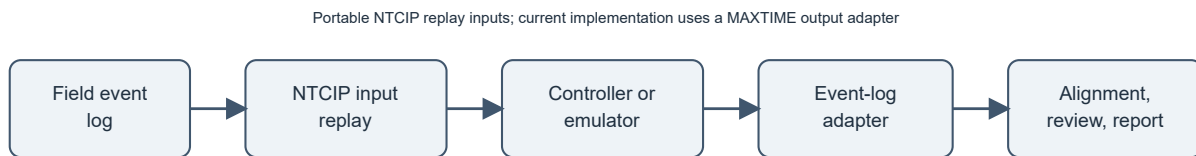


Figure 1: Field-derived testing workflow. Standard NTCIP objects form the replay-input boundary; controller output collection and event normalization require a product adapter.

1 Field event conversion

2 The 25 source logs span approximately 9:00 a.m. to 8:00 a.m. the next day and contain 7,296,218 records, or 574.998
3 configuration-hours. Replay selects vehicle detector, pedestrian detector, and preemption changes. Table 1 shows the
4 conversion implemented in the package.

Input	Event codes	Parameter	NTCIP replay state
Vehicle detector	81 off; 82 on	Detector 1–64	Update one bit in an eight-detector group and write the cumulative integer.
Pedestrian detector	89 off; 90 on	Detector 1–64	Update one bit in an eight-detector group and write the cumulative integer.
Preemption	104 off; 102 on	Preempt number	Write the corresponding Boolean preempt state.

5 Table 1. Conversion from field events to replay inputs.

6 Repeated on/off records are repaired by imputing the missing opposite transition. Dummy detector numbers 65 and
7 above are excluded. Before replay, all calls are reset to zero. The package then sends SNMP SET operations to NTCIP
8 1202 vehicle, pedestrian, and preempt objects at the recorded time of day (American Association of State Highway and
9 Transportation Officials and Institute of Transportation Engineers and National Electrical Manufacturers Association
10 2023). Preserving time of day also exercises coordination and time-of-day plan changes.

11 The evaluated implementation used MAXTIME emulators loaded with production databases. Emulator accommoda-
12 tions removed detector delay/extension, disabled unused input/output modules, removed inverted preempt logic, used
13 localhost peer addresses, and mapped overlap pedestrian calls to dummy detector inputs. Output events were collected
14 through a MAXTIME HTTP interface. Because replay inputs are sent through NTCIP 1202 objects, the input side
15 is not tied to MAXTIME or another controller brand. Applying the method to another controller family requires an
16 output adapter that maps its events to the common timestamp/event/parameter format.

17 Parallel replay and latency control

18 Signals assigned to available emulator targets run concurrently. The 25 configurations were processed in batches
19 over approximately one week without requiring staff to operate each input. After the initial setup, future releases are
20 expected to require only a few hours of staff preparation and review; this is operational experience, not a formal
21 labor study.

22 Replay timing includes a configurable latency offset. During the MAXTIME version 2.18.1 campaigns, the runner
23 also estimated latency from isolated detector-on events and updated the offset during collection. The stored release
24 database contains 707,458 latency samples and 6,577 applied updates. The three campaign databases each contain
25 627,240 replay commands and between 7.57 and 8.02 million controller-output events.

26 OUTPUT ALIGNMENT AND AUTOMATED CHECKS

27 Discrete-event dynamic time warping

28 Events at the same timestamp, or within 0.25 seconds, are grouped into sets of (event identifier, parameter)
29 pairs. Let A_i and B_j be reference and candidate event sets. Their local cost is Jaccard distance:

$$d(A_i, B_j) = 1 - \frac{|A_i \cap B_j|}{|A_i \cup B_j|}.$$

30 Identical event groups cost zero. Missing or additional events increase the cost. The cumulative DTW cost is

$$D(i, j) = d(A_i, B_j) + \min(D(i-1, j), D(i, j-1), D(i-1, j-1)),$$

31 with $D(0, 0) = 0$ and inaccessible borders set to infinity. Backtracking produces an order-preserving path; horizontal
32 and vertical path segments identify additions, omissions, or local shifts rather than hiding them.

Illustrative event grouping before sequence and timing comparison

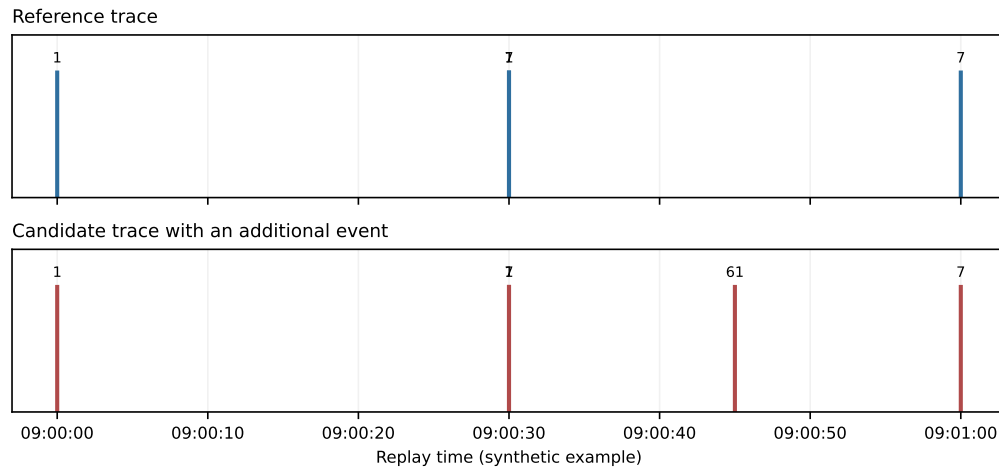


Figure 2: Simplified discrete-event alignment. DTW absorbs small timing shifts while missing and additional event groups retain nonzero costs.

1 Sequence scores are calculated in 45-minute windows advanced every 40 minutes. Sixty seconds are clipped from
2 each window edge to avoid penalizing a cycle split at a boundary. Sequence match is the percentage of aligned event-
3 group pairs with zero Jaccard distance. Timing match is the percentage of exactly matched groups within 0.50 seconds
4 after temporal alignment. A configuration passes at 95 percent sequence match and 90 percent timing match.

5 Vehicle phase-call similarity provides a separate replay-reliability check. Windows below 85 percent are displayed but
6 excluded from the configuration average. This helps distinguish controller differences from a test environment that
7 did not reproduce the intended inputs.

8 **Operational checks and virtual conflict monitor**

9 The automated review is broader than sequence scoring. It compares phase and overlap intervals, clearance durations,
10 pedestrian service, preemption, and coordination-transition states. It highlights short or irregular yellow/red clearances
11 and material changes in occurrence or duration.

12 A software-based virtual conflict monitor reconstructs active phase, overlap, pedestrian, and overlap-pedestrian states
13 from every collected output event. Each state is checked against configuration-specific incompatible pairs. The study
14 definitions contain 863 incompatible pairs across 21 configurations. No conflicts were stored in the MAXTIME 2.15.1,
15 MAXTIME 2.18.1, or modified-overlap-setting campaigns. This automated check is a major increase in test coverage,
16 but it complements rather than replaces a cabinet conflict monitor or formal safety certification.

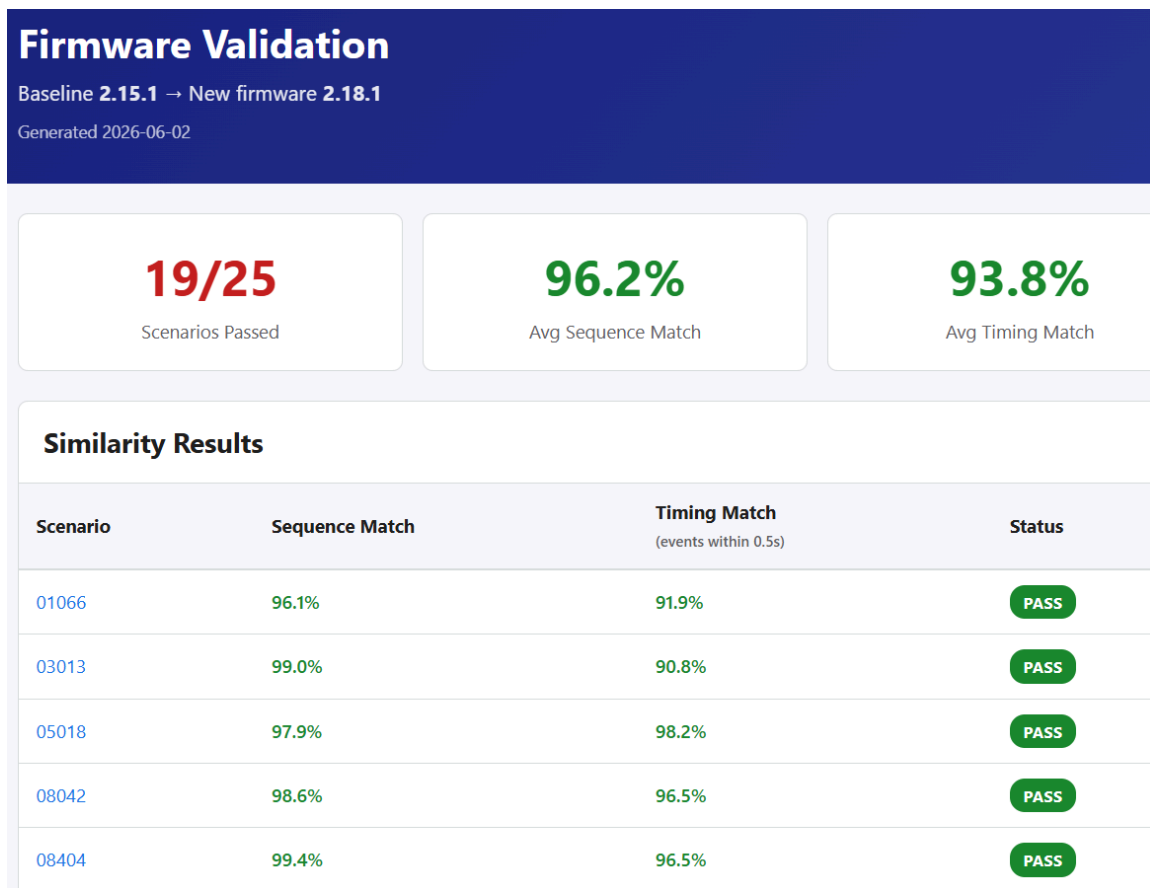


Figure 3: Example automated report. Aggregate sequence and timing results lead to focused views of clearance, pedestrian, preemption, transition, and conflict findings, allowing staff to review exceptions instead of the full event stream.

1 EXPERIMENTAL DESIGN

2 Table 2 separates the two tests. The release comparison represents normal acceptance testing. The same-software
3 comparison provides a repeatability group and a known operational intervention.

Test	Reference	Candidate	Configurations	Question
Software release	MAXTIME 2.15.1	MAXTIME 2.18.1	25	Can full-day outputs be aligned and meaningful re-release differences surfaced?
Overlap clearance settings	MAXTIME 2.18.1	Same software with applicable overlap trail settings	14 changed; 11 unchanged	Are unchanged replays repeatable, and are deliberate changes detected?

4 Table 2. Evaluation design.

5 Occasional truncated overlap yellow and red clearance intervals had been observed. To prevent them, trail-yellow
6 settings were added to applicable overlaps, with trail-red and red-revert settings changed as needed. Transaction
7 histories identified these edits in 14 configurations; 11 had no relevant edit. The test provides a known operational
8 difference while holding the software version constant. A changed setting is detectable only if the recorded day
9 exercises the affected overlap operation.

10 RESULTS

11 Field replay produced repeatable controller outputs

12 All 11 unchanged configurations passed across separate MAXTIME 2.18.1 replay campaigns. Their mean sequence
13 match was 99.6 percent (range 99.1–100.0 percent), and their mean timing match was 97.7 percent (range 90.6–99.7
14 percent). These were independent full-day controller executions, not a recorded output compared with itself. This is

1 the principal feasibility result: the controller responses were deterministic enough for DTW to align and compare them
2 automatically.

3 Eleven of 14 configurations with applicable overlap clearance-setting changes were flagged, while none of the 11
4 unchanged configurations was flagged. The three changed configurations that passed had sequence matches of 99.3,
5 99.9, and 100.0 percent; the recorded input apparently did not materially exercise the affected behavior. Table 3
6 summarizes the result without implying exhaustive coverage of every possible timing change.

Group	Flagged	Passed	Total
Applicable overlap clearance edit	11	3	14
No relevant edit	0	11	11

7 Table 3. Same-software repeatability and parameter-intervention results.

8 The intervention removed short and variable overlap-yellow intervals while otherwise retaining similar operation in
9 the reviewed examples (Figure 4). This shows how the same field trace can evaluate a timing correction as well as a
10 software release.

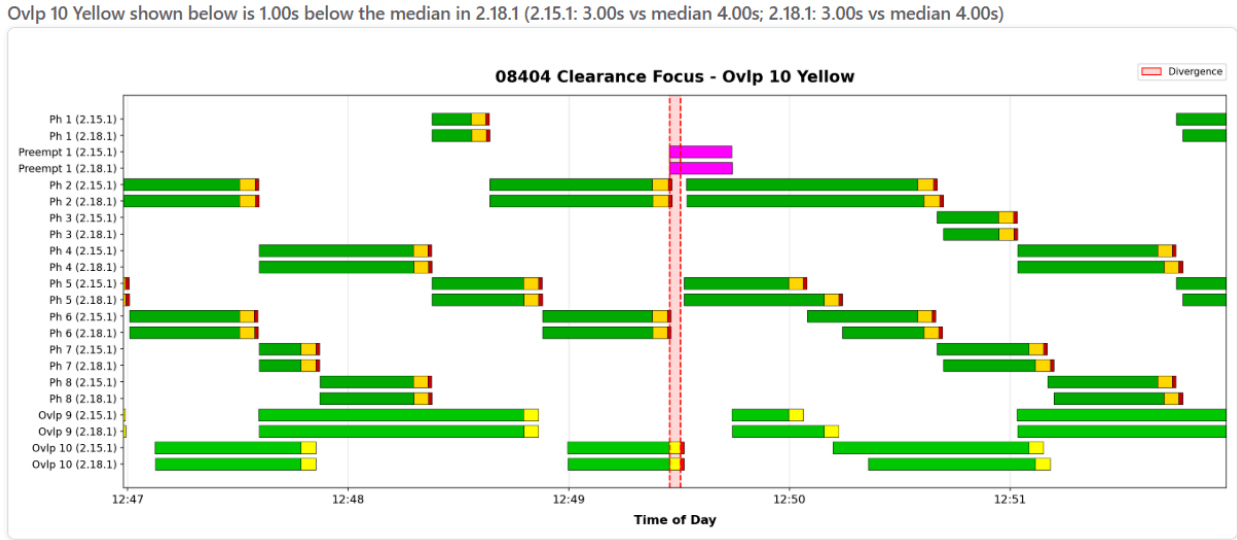


Figure 4: Representative parameter test. Adding overlap trail-yellow and related clearance settings removed short, variable overlap-yellow intervals while the comparison checked the remainder of the replay for unintended changes.

11 Application to software-version acceptance

12 After repeatability was established, the method was applied to acceptance testing of a new software version. Nineteen
13 of 25 configurations passed the aggregate thresholds and six were directed to staff review. Mean sequence match
14 was 96.2 percent and mean timing match was 93.8 percent. Four configurations fell below the sequence threshold;
15 two exceeded the sequence threshold but fell below the timing threshold. These counts describe how the review was
16 organized; the important result is that the aligned traces exposed specific changes in signal operation.

Outcome	Count	Configurations / interpretation
Passed both thresholds	19	Operationally similar overall; detailed diagnostics still identified localized changes.
Sequence review	4	12036, 2B045, 2B094, 2B339. The 12036 result likely reflects peer-to-peer operation not being configured correctly in the test environment.
Timing review	2	2B054 and 2B085.

17 Table 4. MAXTIME version 2.15.1 versus version 2.18.1 release-comparison outcomes.

18 The detailed findings were more informative than the aggregate status:

- 1 • **Rail-preemption fix (13008).** MAXTIME version 2.15.1 could leave Preempt 6 active with the agency's rail-
 - 2 preemption configuration. MAXTIME version 2.18.1 exited correctly (Figure 5). This confirmed a known bug fix.
 - 3 • **Transition improvement (2B049).** MAXTIME version 2.18.1 completed a short-way coordination transition faster,
 - 4 as shown against the programmed split. The trace also raised a separate question: the controller appeared to be in
 - 5 step while still reporting transition (Figure 6).
 - 6 • **Expected clearance behavior (13008).** The report flagged an irregular phase 4 red-clearance interval. Review
 - 7 showed that dynamic red-clear extension was configured, and both versions handled it consistently. The flag
 - 8 demonstrated that an unexpected clearance change would have been visible.
- 9 The first two configurations passed the aggregate thresholds. This is important: the tool did not merely label controllers
- 10 pass or fail. It retained localized differences that confirmed a bug fix and exposed an algorithm change within otherwise
- 11 similar full-day operation.

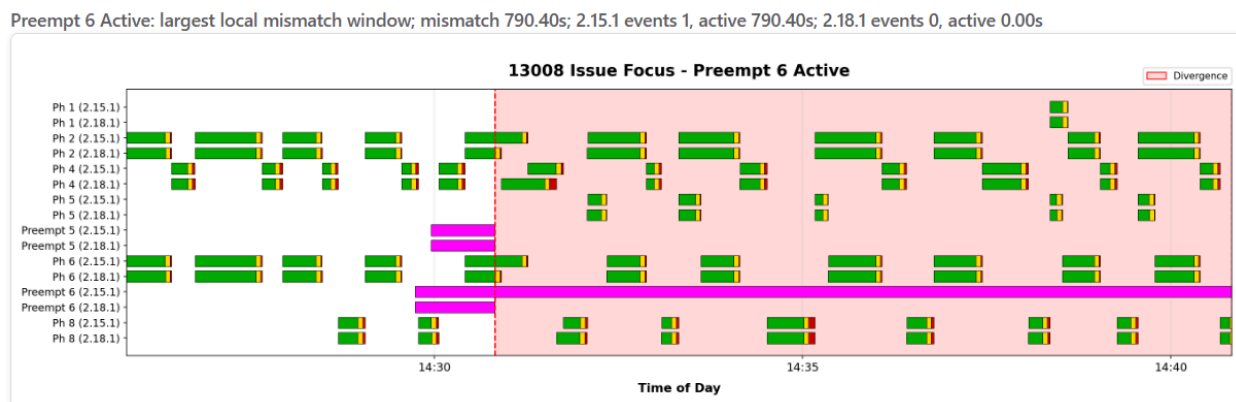


Figure 5: Configuration 13008, Preempt 6. MAXTIME version 2.15.1 remained active under the rail-preemption configuration; version 2.18.1 exited correctly, confirming the known fix.

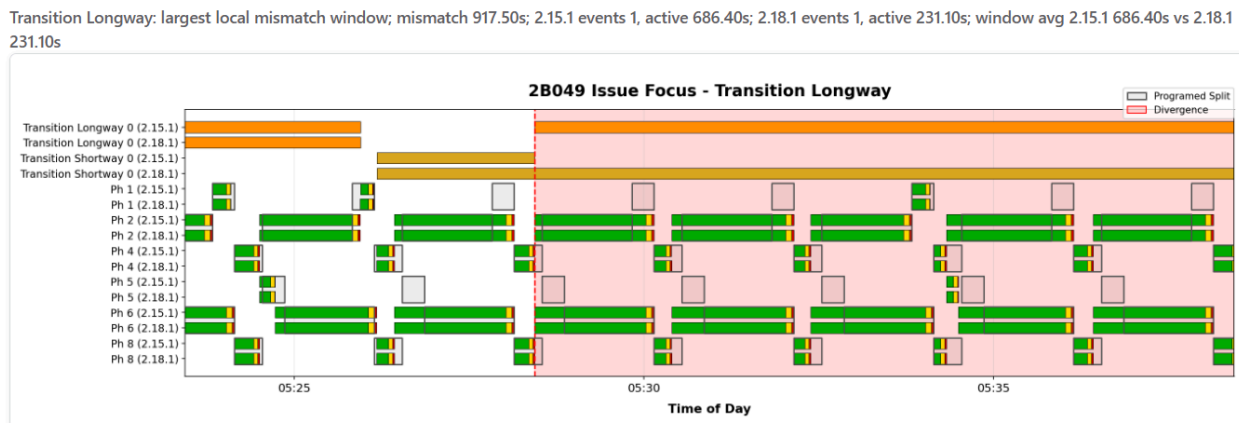


Figure 6: Configuration 2B049, short-way transition. MAXTIME version 2.18.1 returned to the programmed split faster. The aligned trace also showed the controller reporting transition after appearing to be in step.

- 12 No new operation-impacting software defect was confirmed in MAXTIME version 2.18.1 from the reviewed results.
- 13 The new version was otherwise operationally similar across the tested configurations, and the virtual conflict monitor
- 14 recorded no incompatible simultaneous outputs.

15 DISCUSSION

- 16 The method enhances controller acceptance testing; it does not simply automate the same manual test. Staff can still
- 17 check individual functions deliberately, while field replay adds approximately 23 hours of interacting inputs across 25
- 18 actual timing configurations. Available emulators run concurrently, and the software checks millions of output events
- 19 for sequence and timing changes, clearance irregularities, preemption and pedestrian differences, and incompatible

1 signal phases and overlaps. Staff can concentrate on the exceptions instead of choosing and operating every input
2 combination.

3 The high repeatability of the unchanged runs is what makes this practical. If repeated controller outputs could not
4 be aligned, an unexpected software effect would be indistinguishable from ordinary run-to-run variation. Instead, the
5 same inputs produced closely matching sequences, while deliberate timing changes and a software bug fix remained
6 visible. The comparison therefore provides a broad safety net for unforeseen changes in every behavior exercised by
7 the replay. It cannot detect a condition absent from the recorded day, so targeted manual tests and scripted requirements
8 remain necessary.

9 Field logs also capture combinations that are difficult to anticipate: simultaneous vehicle and pedestrian calls, time-of-
10 day plan changes, coordination transitions, and preemption during actual traffic operation. A rare failure or flash event
11 can be saved as a targeted replay, used to reproduce a reported problem, and then run again to test a vendor correction
12 or timing workaround.

13 The accepted baseline is a reference trace, not an assumption that the old operation is always correct. The 13008
14 preemption difference was desirable because the candidate software fixed a baseline defect. Staff must classify each
15 reported difference as expected, beneficial, harmful, caused by the test environment, or unexplained before accepting
16 a new baseline.

17 The package is open source and separates NTCIP inputs from product-specific outputs. The replay-input method is not
18 tied to a controller brand: it can be applied to controllers that implement the required NTCIP objects. A new controller
19 family still needs an output collector, event-code mapping, and validation of timestamp behavior. MAXTIME is the
20 implementation evaluated here, not a limitation of the underlying replay method.

21 **LIMITATIONS**

22 The campaigns used one agency's production configurations and one controller-family emulator. Emulator accommo-
23 dations and incomplete peer-to-peer setup affected generalizability and likely explain the 12036 mismatch. Hardware
24 and a second controller family should be evaluated in future work.

25 A 23-hour trace tests only behavior exercised during that day. Three parameter-modified configurations produced
26 essentially unchanged output, illustrating the need to retain multiple field days and targeted incident traces. The
27 thresholds are engineering review values rather than statistically optimized decision boundaries, and the planned one-
28 factor sensitivity analysis has not yet been run.

29 Five release-threshold review cases remain incompletely classified after accounting for the likely 12036 environment
30 issue. That does not invalidate the feasibility result, but final operational disposition would strengthen the acceptance-
31 test case study. The virtual conflict monitor is also limited to the incompatible pairs defined for 21 configurations and
32 is not a substitute for certified cabinet monitoring.

33 The controller replay requires emulator software, configuration databases, and agency infrastructure. The open-source
34 repository and proposed event bundle can reproduce the comparison and reporting stage without that equipment, but
35 not the original controller execution.

36 **CONCLUSIONS**

37 This study found that field-derived high-resolution event replay is feasible for comprehensive controller testing.
38 Independent unchanged runs averaged 99.6 percent sequence match and 97.7 percent timing match, demonstrating that
39 DTW could reliably align complete controller-output sequences. The comparison then flagged 11 of 14 configurations
40 with deliberate overlap clearance-setting changes and none of the unchanged configurations. Applied to a software
41 update, it surfaced real operational differences within otherwise similar full-day operation, including a confirmed rail-
42 preemption exit bug fix. The virtual conflict monitor found no incompatible signal phases or overlaps in the three
43 campaigns.

44 The significance is the increase in test coverage. Instead of relying only on staff to select and toggle individual
45 inputs, agencies can replay actual field activity across many production configurations and automatically compare the
46 resulting signal phases, clearances, pedestrian service, preemption, coordination, and conflicts. This creates a practical
47 opportunity to discover unforeseen software or timing effects in the test environment rather than after field deployment.
48 The same process can reproduce field failures and verify proposed fixes against identical inputs.

1 The open-source package provides a reusable NTCIP-based input interface and a comparison method that other
2 agencies can adapt. Controller-specific output collection and event mapping are still required, but the successful
3 alignment and detection results establish that the underlying approach works.

4 **ACKNOWLEDGMENTS**

5 The author thanks Chris Primm, State Traffic Operations Engineer, Oregon Department of Transportation, for
6 reviewing the results and contributing ideas for testing.

7 **CODE AND DATA AVAILABILITY**

8 Signal-Replay is an open-source Python package available at [https://github.com/ShawnStrasser/ATC-Signal-](https://github.com/ShawnStrasser/ATC-Signal-Replay)
9 [Replay](https://github.com/ShawnStrasser/ATC-Signal-Replay). It includes replay and comparison code, tests, documentation, and an offline example. The final submission
10 should cite release v0.2.0 or the exact public commit. A comparison-only data bundle can include replay inputs,
11 collected outputs, parameters, and expected results without controller firmware or proprietary configuration databases.

12 **GENERATIVE AI DISCLOSURE**

13 Generative artificial intelligence tools, including Claude Code and OpenAI Codex, were used to assist with software
14 development and review, data-analysis workflows, figure preparation, manuscript review, and document formatting.
15 The author reviewed and tested the software, verified the cited sources and reported results, and accepts responsibility
16 for the research methods, findings, interpretations, and final manuscript.

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